

expected return of commodities is lower than those of stocks even though the price volatility is similar.

Pension fund trustees tend to view risk as the uncertainty that their employee pension obligations will be met in the future. Defined-benefit pension plans are managed so that the future payments to retirees are matched by the future expected value of the pension fund. An actuarial assumption of pension obligations is compared to a forecast plan value based on an estimated return on assets and future contributions. If the forecast plan value is equal to the obligation, the plan is fully funded and no excess contributions are needed from the employer. However, if the forecast value is less than the plan's obligations, there is an underfunded liability, and that represents a financial risk. An underfunded pension obligation means that the employer will have to commit more resources to the plan to maintain its solvency, and this could cause financial difficulty for the employer.

Mutual fund managers see risk as the underperformance of their fund compared to other fund managers with the same investment objective. If a large growth fund manager does not perform well in relation to other large growth fund managers, then the fund will lose assets as investors liquidate shares, not to mention that the manager will likely lose his or her job.

Unlike academics, pension fund trustees, and money managers, individual investors define risk in a more direct way. They define it as losing money. Nothing gets people's attention faster than when the value of their account begins to fall. It does not matter if a person made money for several years prior; it is the here and now that matters.

Consider the year 1987. It was the year the stock market crashed. What was the return of the market that year? The S&P 500 was up by about 36 percent through September 1987, and then it fell apart. On Friday, October 16, stocks unexpectedly fell by 9 percent. The following Monday, known as Black Monday, prices came crashing down another 23 percent. Investors were shell-shocked. However, despite the October collapse, the market still stood in positive territory at the end of the month, and by year-end the S&P 500 was up a respectable 5.1 percent. No investor who had money in a diversified stock portfolio for the entire year in 1987 lost money, but that is not what people remember. We only remember how bad it felt to lose money on Black Monday.